

MediAssist Project Progress & Integration Document

Project Title: MediAssist – AI Powered Smart Medication Management System

Project Objective

MediAssist is an AI-powered healthcare application developed using Flutter, Firebase, Python, Machine Learning and OCR. The application provides AI Prescription Analysis, Medication Reminder System, AI Symptom Analysis, Medical Report Analysis, Doctor Appointment Booking, Personal Health Profile and Secure Cloud Storage. The architecture is modular so every team member can work independently and integrate later.

Current Project Status

- Overall completion: Approximately 60%.
- Authentication module completed.
- Profile module completed.
- Dashboard completed.
- Prescription module completed.
- Reminder and notification system implemented.
- AI Symptom Analyzer in progress.
- OCR integration pending.
- Appointment module pending integration.

Authentication Module

- Firebase Authentication implemented.
- User Registration.
- User Login.
- Persistent Login.
- Logout.
- Splash Screen checks authentication state.

Authentication Flow

```
graph TD
    A[Splash Screen] --> B[Firebase Authentication]
    B --> C[Already Logged In?]
    C -- YES --> D[Home]
    C -- NO --> E[Login]
```

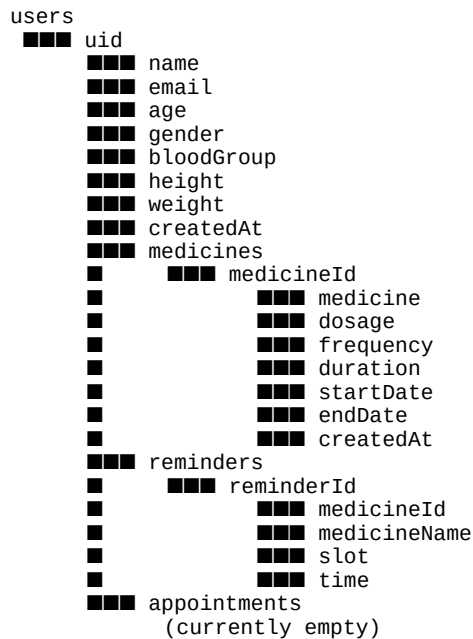
User Profile Module

- Stores Name, Age, Gender, Height, Weight and Blood Group.
- Profile information is stored in Firestore.

Home Dashboard

- Cards available: Prescription Analysis, Medication Reminder, Symptom Analyzer, Medical Reports, Doctor Appointment and Profile.
- Dark professional medical theme implemented.

Firestore Database Structure



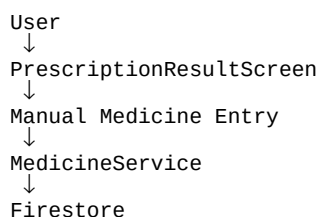
Prescription Module (OCR Integration Point)

Current Flutter screen: PrescriptionResultScreen. It currently supports manual medicine entry. The OCR module must populate the existing fields automatically without redesigning the UI.

Current Prescription UI

- Dynamic medicine cards.
- Medicine Name field.
- Dosage dropdown (250, 500, 650, 1000 mg).
- Frequency dropdown (1-0-0, 1-1-0, 1-0-1, 1-1-1, 0-1-1, 0-0-1).
- Duration dropdown (3,5,7,10,14,30 Days).
- Add Medicine.
- Remove Medicine.

Current Prescription Flow



↓
Reminder Screen

Required OCR Flow

Camera
↓
Prescription Image
↓
OCR
↓
Extract Text
↓
Medicine Parser
↓
Flutter
↓
Auto-fill Medicine Cards
↓
User Review & Edit
↓
Save

Medication Reminder Module

- Morning, Afternoon and Night reminders implemented.
- Time Picker implemented.
- Android local notifications working.
- Persistent login supported.

Reminder Architecture

Medicine
↓
Reminder Screen
↓
Select Time
↓
Generate notifications from Start Date to End Date
↓
Notifications stop automatically after duration

Notification System

- Libraries: flutter_local_notifications, timezone, flutter_timezone.
- Manifest configured with POST_NOTIFICATIONS, SCHEDULE_EXACT_ALARM and RECEIVE_BOOT_COMPLETED.

AI Symptom Analyzer

Planned pipeline: Flutter → Symptom Selection → Flask API → Random Forest Model → Disease Prediction → Description → Medication → Diet → Workout → Precautions → Flutter Result Screen.
Dataset selected: SympScan (100 diseases, 200+ symptoms).

Medical Report Analyzer

Pipeline: Medical Report → OCR → Extract Parameters → AI Summary → Flutter. Will explain Hemoglobin, Blood Sugar, Platelets, WBC and other clinical values.

OCR Team Responsibilities

- Develop OCR and Medicine Parser.

- Return JSON containing medicine, dosage, frequency and duration.
- Do NOT redesign Flutter UI.
- Integrate with existing PrescriptionResultScreen.

Expected OCR JSON

```
[
  {
    "medicine": "Crocin",
    "dosage": "500 mg",
    "frequency": "1-0-1",
    "duration": "5 Days"
  },
  {
    "medicine": "Azithral",
    "dosage": "250 mg",
    "frequency": "0-1-0",
    "duration": "3 Days"
  }
]
```

Appointment Module Responsibilities

- Implement Doctor List.
- Appointment Booking.
- Date & Time Slot.
- Confirmation.
- Firestore Storage.
- Appointment History.
- Appointment Cancellation.
- Use existing appointments collection.

Flutter Folder Structure

```
lib/
  constants/
    theme.dart
  screens/
    login_screen.dart
    register_screen.dart
    splash_screen.dart
    home_screen.dart
    profile_screen.dart
    profile_setup_screen.dart
    prescription_result_screen.dart
    reminder_screen.dart
    main_navigation_screen.dart
  services/
    medicine_service.dart
    reminder_service.dart
    notification_service.dart
  main.dart
```

Coding Standards

- Flutter
- Dark Theme
- Responsive UI
- Material Design

- Firebase Authentication
- Firestore Database
- Modular Architecture
- Service Layer for Firebase.

Current Development Roadmap

- Completed: Authentication, Firebase Integration, Dashboard, Profile, Prescription UI, Medicine Storage, Reminder System, Notifications.
- In Progress: AI Symptom Analyzer.
- Pending: OCR Integration, Medical Report Analyzer, Appointment Module, AI Integration, Final Testing and Performance Optimization.

Important Notes

Do not modify the Firestore schema without discussion. Keep OCR and Appointment modules modular. Return structured APIs/JSON so Flutter integration remains simple. The Prescription UI must remain unchanged; only auto-fill functionality should be added.